

Module/Pathway Review: Bacterial

ImuABC Damage-Induced Mutagenesis in

Pseudomonas putida KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Pathway/bucket query:** error-prone translesion synthesis **Generic module:** SOS-regulated, error-prone DNA synthesis by an ImuA–ImuB–DnaE2 cassette **Local candidate genes supplied:** 0 (empty metadata table — resolved de novo from UniProt/KEGG + literature)

Summary

The ImuABC/DnaE2 damage-induced mutagenesis module is **fully present and satisfiable** in *Pseudomonas putida* KT2440, despite the empty local candidate-gene table. All five expected module steps map to identifiable genes, and the effector portion of the module is organized as a single contiguous SOS operon. The cassette comprises **PP_3116 (lexA2)**, **PP_3117 (imuA)**, **PP_3118 (imuB)**, and **PP_3119 (dnaE2/imuC)**, with the shared upstream damage sensor **RecA (PP_1629)** completing the RecA-dependent signaling step. This assignment is anchored by direct target-organism evidence: a genome-wide microarray study in KT2440 established that one of the strain's two LexA paralogs, **LexA2**, regulates only its own transcriptional unit, which contains *imuA*, *imuB*, and *dnaE2* ([PMID: 17933893](#)).

The most curation-relevant conclusion is that this module must be **specialized to the correct paralogs** in KT2440. The strain carries two LexA repressors: LexA1 (PP_2143) runs the conventional *E. coli*-like SOS regulon, while LexA2 (PP_3116) is a dedicated gate for the *imu* cassette. The module's "LexA-family repression" step should therefore be pinned to **LexA2, not the housekeeping LexA1**. Similarly, the DnaE2 step must be kept distinct from the essential replicative Pol III α -subunit **DnaE1 (PP_1606)**; both share EC 2.7.7.7 but only DnaE2 is the mutagenic translesion enzyme. Functional support for the DnaE2/ImuC step is direct in pseudomonads: alkylation-damage-induced mutagenesis in *P. putida* and *P. aeruginosa* is "largely ImuC-dependent" ([PMID: 28118378](#)).

The two accessory steps — **ImuA (PP_3117)** and **ImuB (PP_3118)** — are covered by operon context and LexA2 co-regulation in the target strain, but their mechanistic roles are transferred by homology from *Caulobacter crescentus* and *Myxococcus xanthus* models; they have not been individually knocked out for mutagenesis in KT2440. These two steps should be marked **covered but candidate_uncertain**. One explicit over-annotation risk requires a curator note: ImuB carries a Y-family (UmuC/IMS) polymerase domain but is expected to be **catalytically dead**, functioning as a scaffold rather than an active polymerase. The cassette is **single-copy**, and KT2440 lacks a chromosomal Pol V (no *umuDC/rulAB/samB*), making DnaE2 the strain's principal SOS-dependent mutagenic polymerase.

Target-Organism Pathway Definition

What the module includes

The module covers **SOS-regulated, error-prone (mutagenic) DNA synthesis** carried out by the ImuA–ImuB–DnaE2 cassette when replication stalls at a damaged template. Concretely, in KT2440 the process is:

1. DNA damage generates ssDNA; **RecA (PP_1629)** forms an activated nucleoprotein filament that acts as the damage sensor and co-protease.
2. Activated RecA promotes autocleavage of the cassette-specific repressor **LexA2 (PP_3116)**, derepressing the operon.
3. **ImuA (PP_3117)** provides a non-catalytic accessory function, in part by modulating RecA-mediated template switching to favor translesion synthesis over recombinational bypass.
4. **ImuB (PP_3118)** acts as a (catalytically inactive) scaffold that recruits and organizes the polymerase at the damaged replication intermediate.
5. **DnaE2/ImuC (PP_3119)** performs damage-tolerant, low-fidelity DNA synthesis across the lesion, producing mutations.

Neighboring pathways to keep separate

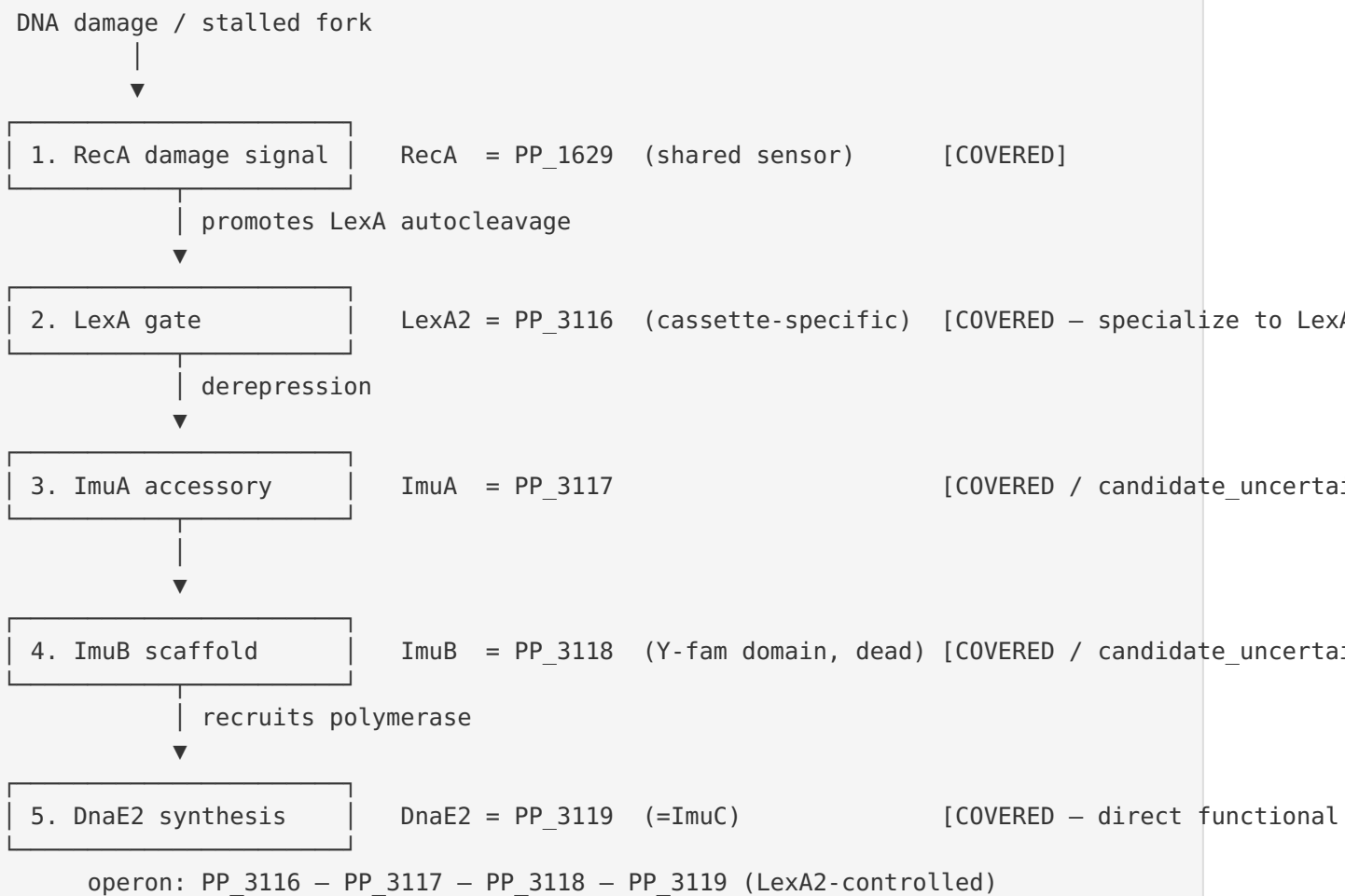
To avoid module bleed, the following processes — all active in KT2440 — must be kept **outside** this module boundary:

Neighboring process	KT2440 genes	Why separate
Canonical <i>E. coli</i> -like SOS regulon	LexA1 (PP_2143)	Different repressor paralog/SOS box; governs recombination/repair genes, not the <i>imu</i> cassette (PMID: 17933893)
Pol IV / DinB mutagenesis	<i>dinB</i> (PP_1203)	Distinct Y-family polymerase; largely RecA-independent , starvation/stationary-phase associated (PMID: 15090515)
Replicative chromosome synthesis	DnaE1 (PP_1606)	Essential Pol III α -subunit; shares EC 2.7.7.7 but is high-fidelity, not damage-induced
Nucleotide excision repair	<i>uvrA/uvrA2/uvrB/uvrC</i>	Error-free excision; separate from mutagenic bypass (PMID: 17720631)
NHEJ / stationary-phase mutagenesis	<i>ligD, ku</i>	Independent mutagenic pathways (PMID: 25942369)
GO/oxidative-damage avoidance	<i>mutY, mutM, mutT, dps</i>	Prevent mutations rather than generate them (PMID: 17545288)

Alternate names and database definitions

- **imuC = dnaE2** — the *P. putida*/*P. aeruginosa* literature increasingly uses **ImuC** for the DnaE2 polymerase ([PMID: 28118378](#)). UniProt for KT2440 uses "Error-prone DNA polymerase" (DNAE2_PSEPK, Q88I82).
- The cassette is variably called **imuAB-dnaE2**, the **imuABC operon**, or the **DnaE2 mutagenesis cassette** across taxa.
- "Error-prone translesion synthesis" (the free-text query) is the umbrella functional description; note it can also encompass DinB/Pol IV and Pol V bypass in other organisms, so the module must be scoped specifically to the ImuAB-DnaE2 machinery.

Expected Step Model and Satisfiability



Step	Expected player	KT2440 gene	Status	Evidence type
1. RecA damage signal	RecA family	PP_1629	covered	Homology + universal SOS role
2. LexA gate	LexA repressor	PP_3116 (LexA2)	covered	Direct KT2440 microarray (PMID: 17933893)
3. ImuA accessory	ImuA family	PP_3117	covered / candidate_uncertain	Operon + regulon context; function by homology
4. ImuB scaffold	ImuB/IMS family	PP_3118	covered / candidate_uncertain	Operon + domain; catalytic-dead by homology
5. DnaE2 synthesis	DnaE2 family	PP_3119 (ImuC)	covered	Direct functional evidence in <i>P. putida</i> (PMID: 28118378)

No step is a gap, and no step is "not expected in the target taxon." The module is satisfiable.

Key Findings — Candidate Genes and Evidence

F001 — The cassette is a contiguous LexA2-controlled operon (PP_3116–PP_3119)

UniProt/KEGG for proteome UP000000556 resolve a contiguous four-gene cluster: **PP_3116** = LexA repressor 2 (LEXA2_PSEPK, P59479); **PP_3117** = ImuA (Q88I84, "Translesion DNA synthesis-associated protein ImuA", 206 aa); **PP_3118** = *imuB* (Q88I83, 472 aa, Y-family-like); **PP_3119** = *dnaE2* (DNAE2_PSEPK, Q88I82, "Error-prone DNA polymerase", EC 2.7.7.7, 1033 aa). Genome-wide microarray analysis in KT2440 showed that "*the other LexA protein (LexA2)*

regulates only its own transcriptional unit, which includes the *imuA*, *imuB*, and *dnaE2* genes" (PMID: 17933893). This is **direct target-strain evidence** for both cassette composition and its repressor, and it satisfies the operon/co-regulation premise of the generic module.

F002 — DnaE2/ImuC (PP_3119) is the principal SOS-dependent mutagenic polymerase

Functional work in *P. aeruginosa* and *P. putida* demonstrated that the TLS polymerase **ImuC (former DnaE2)** confers alkylation-damage tolerance, and that "*mutagenesis induced by MMS in pseudomonads was largely ImuC-dependent*" (PMID: 28118378). This is direct functional evidence in the target species for the DnaE2 synthesis step. It contrasts with **DinB/Pol IV (PP_1203)**, which drives a distinct, largely **RecA-independent**, starvation-associated mutagenic pathway ("*mechanisms different from the classical RecA-dependent SOS response could elevate Pol IV-dependent mutagenesis in starving P. putida cells*", PMID: 15090515). Curators should therefore not conflate DinB-dependent mutagenesis with the ImuABC module.

F003 — Two-LexA architecture: LexA1 (PP_2143) canonical SOS vs LexA2 (PP_3116) *imu* gate

KT2440 encodes two LexA paralogs recognizing distinct SOS-box motifs. The microarray study found that "*one of the two LexA proteins (LexA1) seems to be in control of the conventional Escherichia coli-like SOS response*" (PMID: 17933893), whereas LexA2 governs only the *imuA-imuB-dnaE2* unit (plus a prophage gene, PP_3901). **RecA (PP_1629)** is the shared sensor triggering autocleavage of both repressors. For the module, the LexA step must be assigned to **LexA2 (PP_3116)**; assigning it to LexA1 would be biologically incorrect for this cassette in this strain.

F004 — ImuA (PP_3117) and ImuB (PP_3118) roles inferred by homology

The founding functional study in *Caulobacter crescentus* showed the *imuA-imuB-dnaE2* operon is *recA*-dependent and damage-inducible, and that "*the three genes are required for the error-prone processing of DNA lesions*" (PMID: 15886391). In *Myxococcus xanthus*, "*DnaE2 is an error-prone TLS polymerase, and its functions require ImuA and ImuB*", with ImuA inhibiting RecA-mediated activity to favor TLS (PMID: 34190612). In KT2440, PP_3117 and PP_3118 are supported by operon context and LexA2 co-regulation but have **not** been individually knocked out for mutagenesis; their accessory/scaffold roles are transferred from these related organisms. Transfer is **moderately strong** (conserved operon architecture, conserved regulon) but not experimentally confirmed in the target strain — hence **candidate_uncertain**.

F005 — Domain signatures resolve paralog ambiguity and flag ImuB over-annotation

InterPro/Pfam signatures cleanly separate the players:

Gene	Protein	Key domain signatures	Curation note
PP_3117	ImuA	ImuA_translesion (IPR047610) + SulA/P-loop NTPase fold	RecA-inhibitory fold consistent with Myxococcus ImuA role
PP_3118	ImuB	UmuC / Y-family polymerase (PF00817 IMS, IPR001126)	Expected catalytically dead — do NOT annotate active DNA-pol activity
PP_3119	DnaE2	DnaE2-specific (IPR023073)	Distinct from replicative DnaE1
PP_1606	DnaE1	PolIIIA_DnaE1_PHP (IPR049821)	Essential replicative Pol III α — outside module

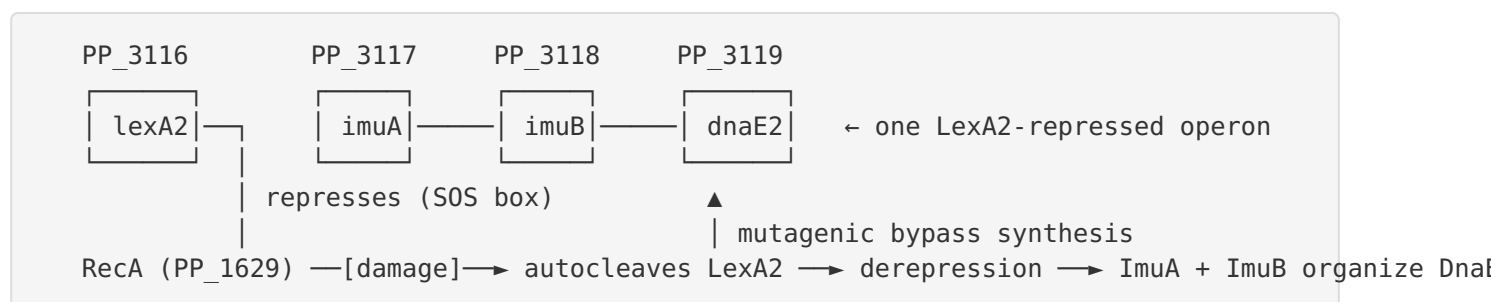
Both DnaE2 and DnaE1 share **EC 2.7.7.7**, so EC number alone cannot distinguish them — the DnaE2-specific InterPro signature (IPR023073) is required.

F006 — Single-copy cassette; no chromosomal Pol V

A proteome-wide InterPro scan of taxon 160488 found: ImuA-family IPR047610 in only PP_3117; DnaE2-specific IPR023073 in only PP_3119 (replicative DnaE1/PP_1606 correctly excluded); and the IMS/UmuC domain PF00817 in only two proteins — ImuB (PP_3118) and DinB/Pol IV (PP_1203). Crucially, **no *umuC/umuD/rulA/rulB/samB* (Pol V) genes are present** on the chromosome. (The *mucA/mucB* hits correspond to the AlgU anti-sigma-factor system PP_1428/PP_1429, not MucAB Pol V.) The module cassette is therefore single-copy, and DnaE2 is the sole SOS-dependent mutagenic polymerase in KT2440 — reinforcing its functional importance. Note that plasmid-borne *rulAB* can be introduced experimentally and does bypass oxidized adenine ([PMID: 17545288](#)), but this is not part of the native KT2440 chromosome.

Mechanistic Model / Interpretation

The module in KT2440 is a clean, self-contained regulatory-effector circuit that can be read directly from genome order:



The circuit's logic: RecA senses ssDNA at a stalled fork and triggers LexA2 self-cleavage; derepression co-expresses the ImuA accessory factor, the ImuB scaffold, and the DnaE2 polymerase in stoichiometric proximity. ImuA biases the outcome away from RecA-mediated recombinational template switching and toward translesion synthesis; ImuB (a catalytically dead Y-family relative) recruits and positions DnaE2 at the primer terminus; DnaE2 then extends across the lesion with low fidelity, generating point mutations. The output is damage-induced, RecA/LexA-gated mutagenesis — mechanistically and regulatorily distinct from KT2440's RecA-independent DinB/Pol IV starvation mutagenesis and from its error-free NER and NHEJ pathways.

Two paralog collisions dominate the curation risk profile and are summarized below:

Collision	Module (correct)	Confounder (wrong)	Discriminator
LexA repressor	LexA2 (PP_3116)	LexA1 (PP_2143)	Regulon target (own operon vs canonical SOS)
DnaE polymerase	DnaE2 (PP_3119)	DnaE1 (PP_1606)	IPR023073 vs IPR049821; both EC 2.7.7.7
Y-family domain	ImuB scaffold (PP_3118)	DinB/Pol IV (PP_1203)	Operon context + catalytic-site degeneracy

Gaps, Ambiguities, and Likely Over-Annotations

- **ImuB polymerase activity (over-annotation risk).** ImuB (PP_3118) carries a UmuC/Y-family polymerase domain and may be automatically annotated with DNA-directed DNA polymerase activity (GO:0003887 / EC 2.7.7.7). Across characterized ImuB proteins, the catalytic aspartates are degenerate and ImuB acts as a **scaffold**, not an active polymerase.

Curators should annotate ImuB with a scaffolding/accessory role and explicitly withhold or qualify polymerase-activity terms.

- **LexA paralog ambiguity.** Automated pipelines may map the module's LexA step to either LexA1 (PP_2143) or LexA2 (PP_3116). Only **LexA2** is correct for this cassette; this is a strain-specific specialization not captured by generic bacterial modules.
- **DnaE1/DnaE2 EC collision.** EC 2.7.7.7 is shared with the essential replicative DnaE1 (PP_1606). EC-based mapping alone would erroneously include DnaE1; the module must rely on the DnaE2-specific signature.
- **ImuA/ImuB functional transfer.** No direct KT2440 knockout data exist for PP_3117 or PP_3118 mutagenesis phenotypes; roles are homology-inferred (candidate_uncertain).
- **DinB cross-talk.** In *P. aeruginosa*, both *dinB* and *imuBC* contribute to ciprofloxacin-resistance mutations ([PMID: 37625357](#)); the two systems can act in parallel, so mutagenesis phenotypes are not uniquely diagnostic of the ImuABC module.
- **Regulatory layering.** The *imu* regulon in *Pseudomonas* also intersects with RpoS/PsrA physiology ([PMID: 30684026](#), [PMID: 11371535](#)); these are context modifiers, not core module steps.

Module and GO-Curation Recommendations

Module step	Recommended status	Assigned gene	Rationale
RecA damage signal	covered	PP_1629	Universal SOS sensor; strong transfer
LexA gate	covered (specialize)	PP_3116 (LexA2)	Direct KT2440 evidence; must not use LexA1
ImuA accessory	covered / candidate_uncertain	PP_3117	Operon+regulon; function by homology
ImuB scaffold	covered / candidate_uncertain	PP_3118	Operon+domain; flag over-annotation
DnaE2 synthesis	covered	PP_3119 (ImuC)	Direct functional evidence in <i>P. putida</i>

Overall module verdict: SATISFIABLE / covered. No step is a gap or `not_expected_in_target_taxon`.

Additional recommendations: - **Module boundary correction for this organism:** the generic module's single "LexA-family repressor" node should be documented as **LexA2-specific** in *Pseudomonas putida*, with a note that LexA1 runs a parallel canonical SOS regulon. This is a `module_needs_revision`-style annotation at the documentation level (paralog specialization), not a structural failure. - **GO annotation guidance:** annotate PP_3118 (ImuB) with an accessory/scaffold function (e.g., DNA damage response / SOS mutagenesis) and **avoid** propagating DNA polymerase catalytic activity. Annotate PP_3119 with translesion synthesis / error-prone DNA polymerase activity (GO:0003887 with an SOS/translesion qualifier), distinguished from the housekeeping replicative activity of PP_1606. - **No new GO term request is strictly required**, but a curator-facing note distinguishing DnaE2 (mutagenic) from DnaE1 (replicative) under EC 2.7.7.7 would prevent recurrent mis-mapping.

Genes to Promote to Full `fetch-gene` Review

1. **PP_3116 (LexA2)** — high priority. Confirm SOS-box binding specificity and its exclusive control of the *imu* cassette; anchor the paralog specialization.
 2. **PP_3118 (imuB)** — high priority. Verify catalytic-dead status and scaffold role; primary over-annotation risk in the module.
 3. **PP_3119 (dnaE2/imuC)** — medium priority. Confirm DnaE2-specific signature and functional annotation; the direct functional evidence already exists but the EC collision with DnaE1 warrants an explicit review note.
 4. **PP_3117 (imuA)** — medium priority. Confirm ImuA_translesion + Sula/NTPase fold and candidate_uncertain functional status.
 5. **PP_1629 (recA)** — low priority. Standard confirmation of the shared sensor.
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Evidence Base

PMID	Organism / scope	How it supports or challenges the review
17933893	Direct KT2440	LexA2 regulates the <i>imuA-imuB-dnaE2</i> unit; LexA1 runs canonical SOS. Anchors cassette composition, repressor identity, and paralog specialization.
28118378	Direct <i>P. putida</i>/P. <i>aeruginosa</i>	MMS-induced mutagenesis is largely ImuC(DnaE2)-dependent — direct functional support for the DnaE2 step.
15886391	<i>C. crescentus</i>	Founding study: all three <i>imuA-imuB-dnaE2</i> genes required for error-prone lesion processing — basis for ImuA/ImuB role transfer.
34190612	<i>M. xanthus</i>	DnaE2 requires ImuA and ImuB; ImuA inhibits RecA-mediated activity to favor TLS — supports accessory/non-catalytic roles.
15090515	<i>P. putida</i>	Pol IV/DinB mutagenesis is RecA-independent — distinguishes DinB from the ImuABC module.
37625357	<i>P. aeruginosa</i>	DinB and ImuBC both contribute to ciprofloxacin-resistance mutations — flags parallel-pathway cross-talk.
17545288	<i>P. putida</i>	Plasmid <i>ruLAB</i> (Pol V homolog) bypasses oxidized adenine — clarifies that native chromosome lacks Pol V.
17720631	<i>P. putida</i>	NER dual role in mutagenesis — neighboring pathway to keep separate.
25942369	<i>P. putida</i>	NHEJ (LigD/Ku) stationary-phase mutagenesis — separate module.
30684026 / 11371535	<i>Pseudomonas</i>	RpoS/PsrA regulatory layering on <i>lexA</i> — context modifier, not a core step.

Limitations and Knowledge Gaps

- **No KT2440-specific knockouts** of *imuA* (PP_3117) or *imuB* (PP_3118) for mutagenesis; their roles rest on operon context plus *Caulobacter/Myxococcus* homology.
 - **DnaE2/ImuC functional data** in pseudomonads derive substantially from *P. aeruginosa* and aggregate "pseudomonad" experiments; while *P. putida* is included, strain-KT2440-resolved lesion-bypass biochemistry is limited.
 - **Regulatory detail** of LexA2 autocleavage kinetics and RecA co-protease specificity in KT2440 is inferred, not measured.
 - **Analysis is annotation-/literature-based** (UniProt, KEGG, InterPro, PubMed); no primary sequence realignment or structural modeling was performed to independently confirm ImuB catalytic-site degeneracy.
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Proposed Follow-up Experiments / Actions

1. **Targeted knockouts in KT2440:** $\Delta imuA$, $\Delta imuB$, $\Delta dnaE2$ single mutants; assay MMS/UV-induced and ciprofloxacin-resistance mutation frequencies to confirm each step's requirement directly in the target strain.
 2. **ImuB catalytic-site mutagenesis:** align PP_3118 against active UmuC/DinB polymerases to verify degenerate catalytic aspartates; test a catalytic-restoration or catalytic-dead point mutant to confirm the scaffold model.
 3. **LexA2 SOS-box footprinting:** EMSA/ChIP to confirm LexA2 binds the *imu* operon promoter and to delimit the LexA2 regulon versus LexA1 in KT2440.
 4. **Epistasis with DinB:** $\Delta dinB \Delta dnaE2$ double mutants to partition SOS-dependent (ImuABC) from RecA-independent (Pol IV) mutagenesis.
 5. **Curation action:** update the module document to specialize the LexA node to LexA2, add the ImuB over-annotation caveat, and add the DnaE1/DnaE2 EC-2.7.7.7 disambiguation note.
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Report generated from a 3-iteration autonomous curation review. Evidence tiers: direct KT2440 (LexA2 regulon, cassette composition); direct pseudomonad (DnaE2/ImuC mutagenesis); homology-transferred (ImuA/ImuB mechanistic roles from Caulobacter/Myxococcus).

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